

TYNDA, Russian Federation

WMO: 304990

Lat: 55.154N

Lon: 124.726E

Elev: 528

StdP: 95.14

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-40.2	-38.1	-42.7	0.1	-39.9	-40.7	0.1	-37.8	5.5	-28.4	5.2	-28.7	1.9	250	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	29.0	18.4	27.0	17.7	25.1	16.9	20.2	25.8	19.2	24.2	18.2	22.9	1.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.3	22.0	17.5	13.4	21.2	16.5	12.5	20.4	60.7	25.7	57.3	24.2	53.9	23.1	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.3	3.8	-42.9	32.2	2.6	1.5	-44.7	33.3	-46.2	34.2	-47.7	35.0	-49.6	36.1		
(5)			-42.8	22.0	2.3	1.6	-44.5	23.1	-45.8	24.0	-47.1	24.9	-48.8	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.3	-27.7	-22.5	-12.9	-1.3	7.8	15.4	18.0	15.3	7.2	-4.0	-19.5	-28.0		
	DBStd	17.33	5.18	5.05	6.05	5.02	4.00	3.58	3.02	3.57	4.24	6.03	6.87	5.34		
	HDD10.0	5831	1169	911	709	339	92	4	0	4	104	435	886	1178		
	HDD18.3	8301	1427	1144	967	589	327	99	42	106	334	693	1136	1437		
	CDD10.0	627	0	0	0	1	23	167	249	168	20	0	0	0		
	CDD18.3	55	0	0	0	0	0	12	32	11	0	0	0	0		
	CDH23.3	829	0	0	0	0	16	252	401	158	3	0	0	0		
	CDH26.7	201	0	0	0	0	1	57	111	32	0	0	0	0		
	Wind	WSAvg	1.4	2.0	1.7	1.7	1.5	1.4	1.0	0.9	1.2	1.4	1.6	2.0		
(15) Precipitation	PrecAvg	607	8	7	10	28	59	112	136	125	65	34	19	9		
	PrecMax	972	25	16	35	87	111	533	304	271	150	90	55	29		
	PrecMin	419	1	0	0	2	20	10	14	16	17	4	4	1		
	PrecStd	127	6	5	9	18	23	97	65	59	34	21	11	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.5	-5.0	5.5	18.1	25.8	30.8	31.9	29.7	23.2	12.6	-0.7	-11.4		
		MCWB	-13.7	-7.6	0.5	8.8	13.9	17.7	19.7	18.9	15.2	6.5	-3.2	-12.6		
	2%	DB	-15.2	-8.6	2.4	13.5	21.9	28.2	29.7	27.2	19.6	9.2	-4.2	-14.5		
		MCWB	-16.0	-10.3	-1.6	6.0	11.4	16.9	19.5	18.2	13.2	4.5	-5.3	-15.3		
	5%	DB	-17.5	-11.5	-0.3	9.8	19.2	26.3	27.5	24.9	17.1	6.5	-6.5	-17.5		
		MCWB	-18.2	-12.8	-3.5	3.7	10.2	16.3	18.5	17.6	11.7	2.6	-7.6	-18.1		
	10%	DB	-19.6	-14.0	-2.5	7.4	16.6	24.2	25.4	22.6	14.9	4.1	-9.1	-20.0		
		MCWB	-20.1	-15.1	-5.0	2.2	9.0	15.4	17.9	16.8	10.3	1.0	-10.1	-20.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-13.7	-7.5	0.7	9.0	14.4	20.2	21.9	20.7	16.1	7.0	-3.1	-12.4		
		MCDB	-12.7	-5.5	5.4	17.2	23.9	26.6	28.6	25.3	21.0	11.3	-1.1	-11.5		
	2%	WB	-16.0	-10.4	-1.6	6.0	12.6	18.6	20.7	19.5	14.3	4.9	-5.4	-15.3		
		MCDB	-15.2	-8.8	2.0	13.1	20.0	25.0	26.7	24.4	18.0	8.9	-4.2	-14.5		
	5%	WB	-18.2	-12.8	-3.3	4.0	11.0	17.4	19.7	18.5	12.7	2.9	-7.6	-18.1		
		MCDB	-17.6	-11.6	-0.5	9.4	17.5	24.0	25.1	23.2	15.6	6.1	-6.5	-17.6		
	10%	WB	-20.1	-15.1	-5.1	2.4	9.5	16.3	18.9	17.4	11.1	1.2	-10.1	-20.4		
		MCDB	-19.6	-14.1	-2.5	6.8	15.4	22.4	23.4	21.7	14.3	3.9	-9.1	-20.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	13.6	14.0	12.1	12.5	13.7	12.0	12.4	11.6	10.5	10.7	8.7		
		MCDBR	10.5	15.0	15.3	16.1	18.5	18.3	17.1	17.0	16.0	13.1	11.1	9.0		
		MCWBR	9.9	13.4	11.7	9.7	9.9	8.1	7.8	8.6	9.4	8.7	9.6	8.5		
	5% WB	MCDBR	10.5	14.9	14.5	15.0	16.4	15.0	13.4	13.3	12.7	12.5	10.8	9.0		
		MCWBR	9.8	13.4	11.2	9.3	9.1	7.5	6.7	7.2	8.1	8.5	9.4	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.240	0.271	0.313	0.426	0.485	0.458	0.461	0.399	0.346	0.315	0.262	0.236		
	taud		2.119	2.104	2.044	1.863	1.926	2.124	2.157	2.327	2.416	2.292	2.121	2.053		
	Ebn at Noon		694	800	844	781	759	791	780	813	816	743	643	589		
	Edh at Noon		61	91	124	175	175	145	138	110	87	76	60	51		
(44) All-Sky Solar Radiation	RadAvg		1.06	2.20	4.01	5.47	5.07	5.67	4.98	4.26	3.15	2.03	1.22	0.75		
	RadStd		0.04	0.07	0.14	0.32	0.36	0.47	0.50	0.45	0.28	0.14	0.07	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page